

A strict Rayleigh bound state in the continuum in a two-groove underwater acoustic metagrating

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At a Rayleigh threshold, zero normal flux does not by itself imply a bound state: a finite grazing pressure remains non-square-integrable. We construct a strict Rayleigh-threshold bound state in the continuum in a rigid, two-groove underwater acoustic metagrating, with both groove widths exceeding 1 mm. At 200 kHz in water, the state occurs at $(\kappa, \Omega) = (0.083848696553, 0.916151303447)$, or $\theta = \pm 5.251216^\circ$. For positive angle the finite-flux $n = 0$ amplitude and the grazing $n = -1$ amplitude both vanish. The reduced homogeneous operator has singular-value ratio 1.26×10^{-15} , while an unconstrained full-operator calculation at the same endpoint gives 7.72×10^{-16} . A same-geometry complex-pole continuation reaches the endpoint with a local finite-model $Q \sim |\Delta\kappa|^{-1.996}$ trend. Homogeneous fields verify exterior localization, and two independent aperture phasors cancel the two radiation constraints. A single large rectangular cavity remains at residual 0.125 within the searched family. Random dimensional perturbations show that the strict cancellation is fabrication sensitive. All results refer to an infinite, rigid, lossless modal model.

A bound state in the continuum (BIC) is a localized eigenstate at a real frequency embedded in propagating radiation channels. It is defined by the vanishing coupling of a *homogeneous* mode to every radiative channel, not by a dip in a driven reflectance spectrum [1–3]. Symmetry, destructive interference, and topological charge provide established routes to photonic BICs [4, 5]; trapped-light observations and perturbative pole analyses show how leaky resonances approach such states [6–8].

A diffraction threshold requires stricter bookkeeping. When a Floquet order opens, its normal wavenumber and normal power flux vanish, but a nonzero pressure amplitude is independent of the surface-normal coordinate and is therefore not square-integrable. The singular structure of scattering matrices at channel openings is well known [9, 10]; at such a point, a strict BIC must cancel the threshold *amplitude* as well as every finite-flux open amplitude. The recent electromagnetic Rayleigh-BIC construction frames this event as a resonant pole reaching a Rayleigh branch point [11].

Acoustic metagratings offer compact, independently tunable cavity modes [12–15]. Waterborne implementations have demonstrated anomalous and negative-order reflection [16–18], and underwater resonant systems have exhibited high- Q and merged-BIC behavior [19, 20]. Here we ask a narrower question: can an acoustic grating at a single Rayleigh opening support a square-integrable homogeneous null when one additional diffraction order is already propagating? We answer this question constructively and test the threshold coefficient directly.

Consider a sound-hard surface of period a occupying $y \leq 0$, with water in $y > 0$ and time dependence $e^{+i\omega t}$.

For incidence toward the surface and outgoing reflection,

$$p_{\text{inc}} = e^{-ik_{x,0}x + ik_{y,0}y}, \quad p_{\text{sc}} = \sum_{n \in \mathbb{Z}} A_n^{\text{dr}} e^{-ik_{x,n}x - ik_{y,n}y}, \quad (1)$$

$$k_{x,n} = k_0 \sin \theta + \frac{2\pi n}{a}, \quad k_{y,n} = \sqrt{k_0^2 - k_{x,n}^2}. \quad (2)$$

The outgoing branch has $\Re k_{y,n} \geq 0$ and $\Im k_{y,n} \leq 0$. We use

$$\Omega = \frac{af}{c}, \quad \kappa = \frac{ak_0 \sin \theta}{2\pi} = \Omega \sin \theta, \quad |\kappa + n| = \Omega \quad (3)$$

for a Rayleigh threshold of order n .

The optimized cell in Fig. 1 has

$$\begin{aligned} a &= 6.871134776 \text{ mm}, & g &= 0.926658264 \text{ mm}, \\ (d_1, d_2) &= (4.541418182, 1.080580298) \text{ mm}, \\ (w_1, w_2) &= (3.964770159, 1.053048089) \text{ mm}. \end{aligned} \quad (4)$$

At $f = 200$ kHz and $c = 1500$ m/s,

$$\begin{aligned} \kappa_{\text{BIC}} &= 0.083848696553, & \Omega_{\text{BIC}} &= 0.916151303447, \\ \theta_{\text{BIC}} &= \pm 5.251216^\circ. \end{aligned} \quad (5)$$

For $+\theta$, $n = 0$ is propagating, $n = -1$ is at threshold because $\kappa + \Omega = 1$, and $n = +1$ is evanescent. Reversing the Bloch number exchanges $n = -1$ and $n = +1$.

Inside groove ℓ , a bottom-rigid cosine expansion is retained. Projecting pressure and normal-velocity continuity without eliminating the groove coefficients gives the pole-free homogeneous system

$$F(\omega, \kappa; \mathbf{g}) \begin{bmatrix} \mathbf{A}^{\text{hom}} \\ \mathbf{C} \end{bmatrix} = 0, \quad \mathbf{g} = (d_1, d_2, w_1, w_2, g). \quad (6)$$

The full block matrix, scaling, and driven reduction are given in the Supplemental Material. This separation is

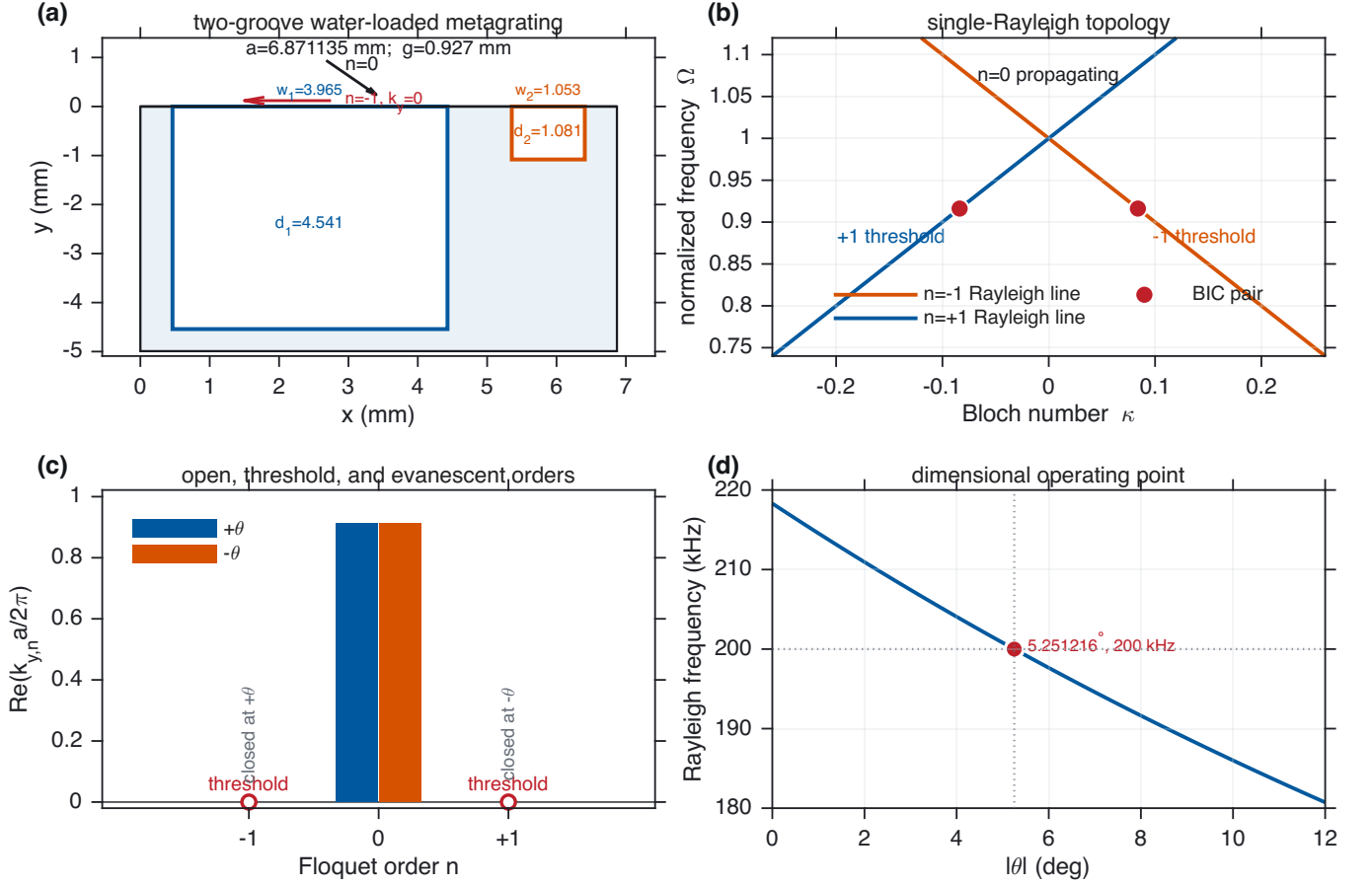


FIG. 1. Geometry and channel topology. (a) Final two-groove cell; dimensions are in millimeters. (b) The two first-order Rayleigh lines and the reciprocal BIC pair. The zeroth order is propagating at both points. (c) Normal wavenumbers for the three displayed orders: for $+\theta$ ($-\theta$), $n = -1$ ($n = +1$) is grazing and the opposite first order is evanescent. (d) The positive-angle point on the dimensional $n = -1$ Rayleigh curve.

essential: A_n^{dr} contains the forced background response, whereas A_n^{hom} belongs to an outgoing eigenmode.

For a single homogeneous exterior order, define the height-integrated pressure norm $\mathcal{N}_n(H)$ by

$$\mathcal{N}_n(H) = \int_0^H |p_n^{\text{hom}}|^2 dy. \quad (7)$$

It evaluates to

$$\mathcal{N}_n(H) = \begin{cases} H |A_n^{\text{hom}}|^2, & k_{y,n} = 0, \\ \frac{|A_n^{\text{hom}}|^2}{2\alpha_n} (1 - e^{-2\alpha_n H}), & k_{y,n} = -i\alpha_n. \end{cases} \quad (8)$$

Thus the positive-angle state is strict only if

$$A_{-1}^{\text{hom}} = 0, \quad A_0^{\text{hom}} = 0. \quad (9)$$

We first impose Eq. (9) by removing those two amplitude columns from the unknown vector, row- and column-scale the remaining operator, and reconstruct its smallest right singular vector. At $(N, K) = (313, 39)$, where N is the

number of exterior orders and K is the cosine-mode count per groove,

$$\begin{aligned} \sigma_{\min}/\sigma_{\max} &= 1.2603643 \times 10^{-15}, \\ r_{\text{raw}} &= 1.3765668 \times 10^{-13}. \end{aligned} \quad (10)$$

This constrained null is not the sole evidence. Evaluating the complete complex-frequency operator at the same final geometry, with the selected $k_{y,-1}$ set exactly to zero, gives $\sigma_{\min}/\sigma_{\max} = 7.7151 \times 10^{-16}$ and an unconstrained null whose open and threshold amplitudes vanish at machine precision.

The solution persists as a reoptimized sequence through $(N, K) = (377, 47)$; all strict and full-operator residuals remain at numerical precision and the maximum normalized parameter step falls to 7.77×10^{-6} . This is evidence for a converging modal sequence, not a formal continuum theorem. A complex-pole continuation performed at the *same* final $(313, 39)$ geometry stays on one branch with minimum successive mode overlap 0.9999995 and reaches $Q = 2.026 \times 10^{12}$ at $|\Delta\kappa| = 10^{-5}$. The resolved local fit is $Q \propto |\Delta\kappa|^{-1.9959}$ [Fig. 2(b)]; we do not

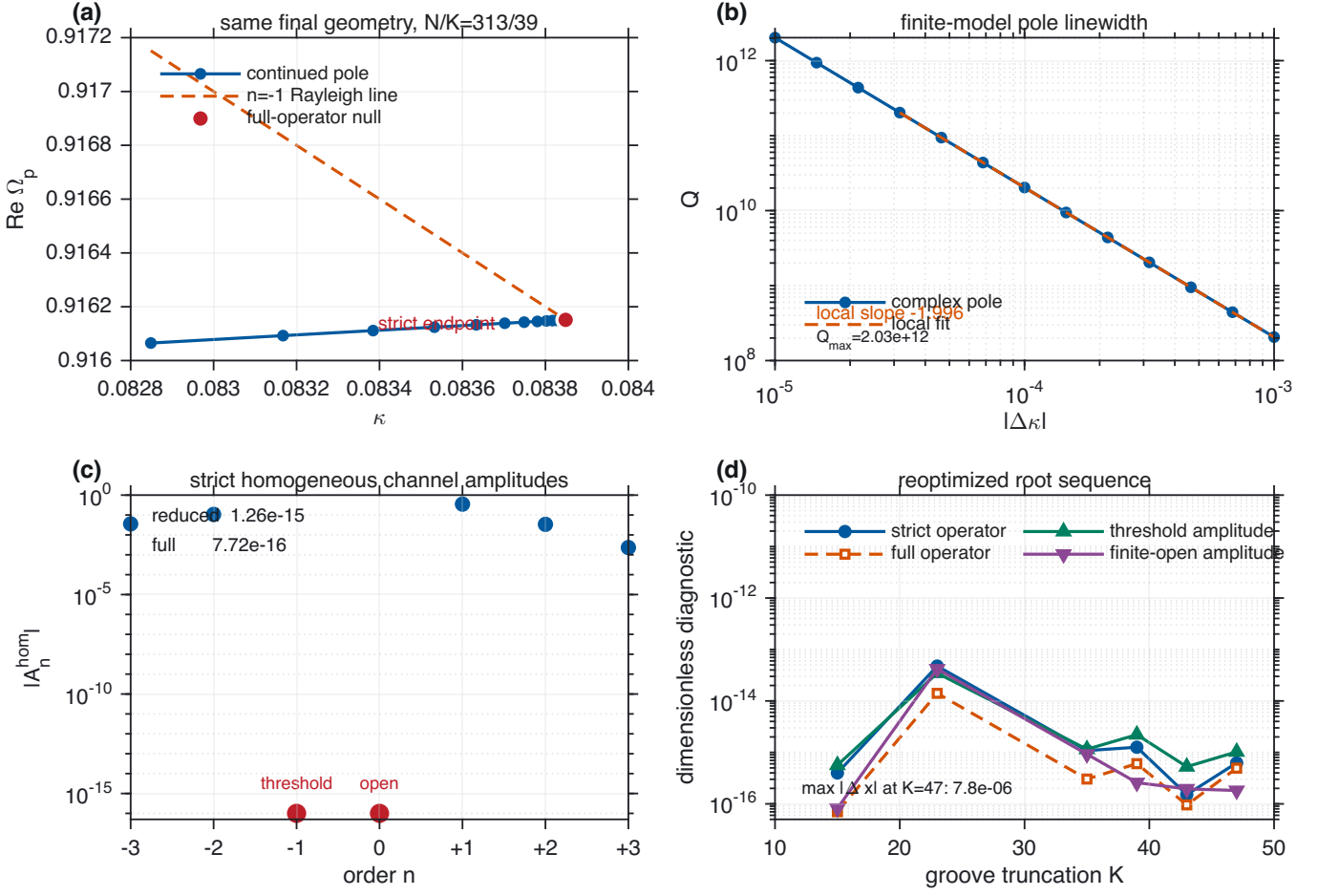


FIG. 2. Strict-null evidence. (a) Final-geometry pole trajectory and the $n = -1$ Rayleigh line. The red endpoint is also a null of the full (313, 39) operator. (b) Pole quality factor and local fit; the exponent is a resolved finite-model trend. (c) Final homogeneous Floquet amplitudes. The finite-flux $n = 0$ and threshold $n = -1$ coefficients vanish, while the displayed remaining orders are evanescent. (d) Reoptimized root sequence: reduced and full singular-value ratios and the independently extracted threshold/open amplitudes remain at machine precision through (377, 47).

assign this finite-model exponent universal significance.

Figure 3 compares homogeneous fields using a common spatial grid and normalization by the maximum pressure inside the grooves. The strict map is reconstructed from the high-order mode, with coefficients truncated to (121, 15) only for stable field rendering. The near-BIC and ordinary-resonance panels are explicitly real-axis smallest-singular-vector surrogates, not exact real-frequency outgoing eigenstates. The strict field contains only evanescent exterior orders: its exterior norm saturates by a few periods, whereas restoring any nonzero grazing amplitude produces the linear term in Eq. (8). At the strict point both normalized channel errors are zero; for the near-BIC surrogate they are $|A_{-1}|/||\mathbf{A}|| = 2.03 \times 10^{-2}$ and $|A_0|/||\mathbf{A}|| = 2.62 \times 10^{-3}$, and the ordinary-resonance surrogate is strongly radiative.

The two zero conditions can be resolved into aperture-

source phasors,

$$A_m^{\text{hom}} = \sum_{\ell=1}^2 \sum_q \mathcal{R}_{m,\ell q} C_{\ell q} = 0, \quad m \in \{-1, 0\}. \quad (11)$$

The two grooves close the $n = -1$ and $n = 0$ phasor polygons separately, with normalized closure errors 1.6×10^{-15} and 1.1×10^{-15} . This motivates, but does not prove universally, the need for two independent phase controls. A global search over a *single large rectangular* groove, retaining transverse modes and varying $0.04 \leq \kappa \leq 0.25$, $0.04 \leq d/a \leq 2$, and $0.50 \leq w/a \leq 0.95$, reaches only $\sigma_{\min}/\sigma_{\max} = 0.124588$ at (221, 44). The comparison is therefore a bounded ablation within this declared shape family, not a no-go theorem for arbitrary one-cavity structures.

The cancellation is also sensitive to fabrication. We perturb (d_1, d_2, w_1, w_2, g) uniformly within $\pm 1\%$, $\pm 2\%$, and $\pm 5\%$ at the fixed Rayleigh point. A deterministic ensemble contains 100 samples per level for a bal-

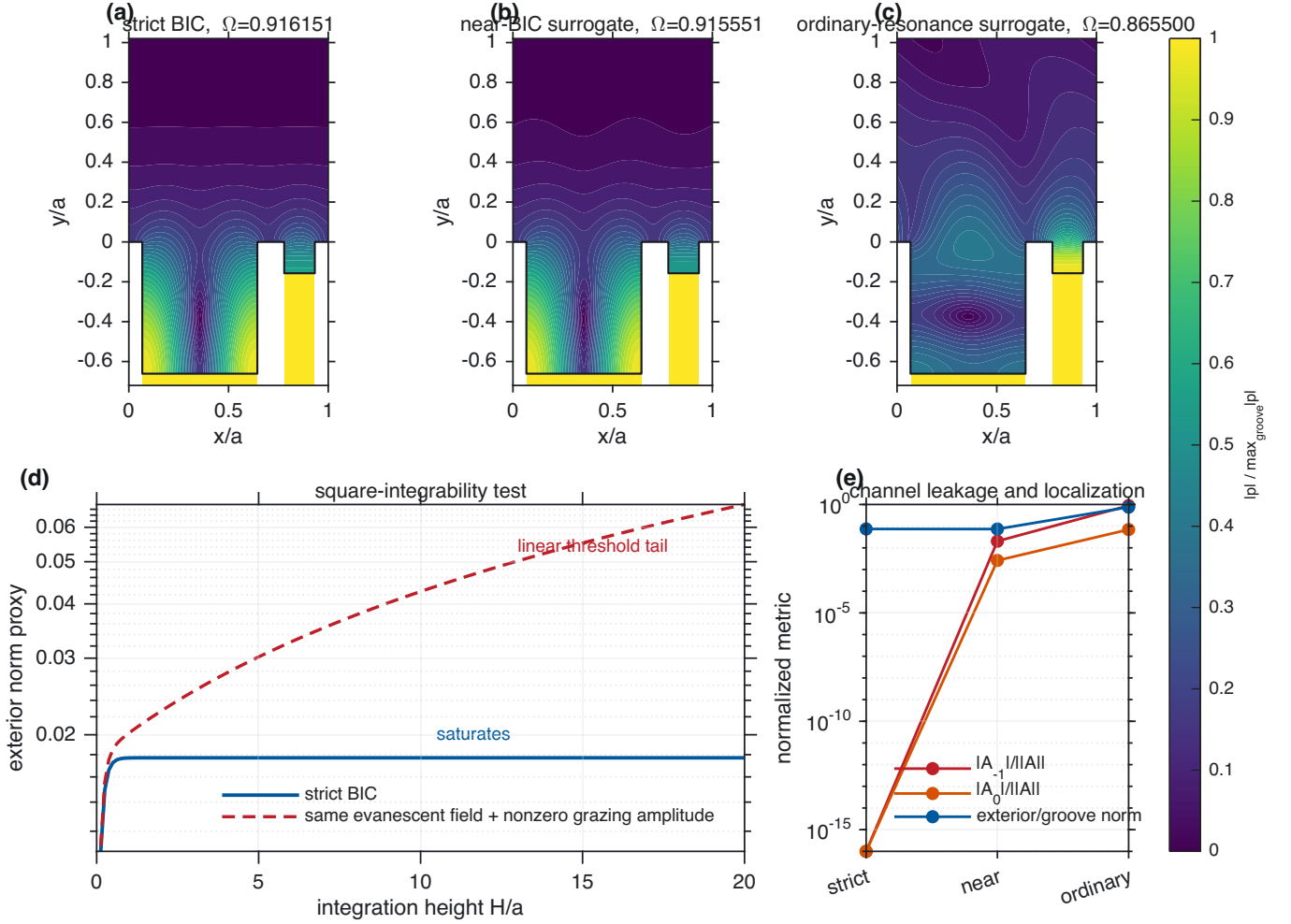


FIG. 3. Homogeneous pressure fields and localization. (a) Strict BIC. (b) Near-BIC real-axis surrogate at $\Omega = 0.9155513$. (c) Ordinary-resonance real-axis surrogate at $\Omega = 0.8655$. All maps show $|p|/\max_{\text{groove}}|p|$ on the same scale; white regions are rigid material. (d) Exterior norm versus integration height for the strict mode and for the same evanescent field with a nonzero grazing component added. (e) Threshold amplitude, finite-open amplitude, and exterior-to-groove norm proxies. Panels (b,c) are diagnostic surrogates, not exact quasinormal-mode fields.

anced one-parameter perturbation and 100 for joint five-parameter perturbations, evaluated at (185, 23). The unperturbed discretization floor is $r_{\text{raw}} = 2.22 \times 10^{-3}$. For one-parameter perturbations the median residual is 0.0136, 0.0359, and 0.0696; for joint perturbations it is 0.351, 0.580, and 0.576. The threshold error in Fig. 4 is extracted from the *unconstrained* full-operator vector, because the strict operator sets it to zero by construction. These statistics quantify sensitivity and should not be read as fabrication acceptance probabilities.

The ideal equations are dimensionally similar. If every length is scaled by s while normalized parameters are fixed,

$$f_{\text{BIC}}(c, s) = \frac{\Omega_{\text{BIC}} c}{s a_0} = f_0 \frac{c}{c_0} \frac{1}{s}. \quad (12)$$

Direct operator evaluations preserve the null for $c = 1400$ – 1600 m/s when the frequency is retuned from

186.667 to 213.333 kHz, and for $s = 0.5$ – 2 when the frequency is retuned from 400 to 100 kHz. This is conditional similarity, not environmental robustness of a fixed device: at fixed geometry, 200 kHz, and angle, changing c moves the structure away from the Rayleigh point. Full sweeps are reported in the Supplemental Material.

Driven scattering is likewise kept separate from the eigenstate proof. A direct fixed-geometry calculation at $\theta = \pm 6^\circ$ and $f = 200.216874$ kHz exchanges $n = -1$ and $n = +1$ under angle reversal and obeys reciprocity, but its target efficiency is strongly truncation dependent (0.3061 at (313, 39) and 0.0572 at (401, 50)). We therefore do not claim a converged one-way device. A distinct off-Rayleigh point reaches 0.9996185 target efficiency at (401, 50); it is an ordinary anomalous reflector and is not BIC-enhanced. Both audits are relegated to the Supplemental Material.

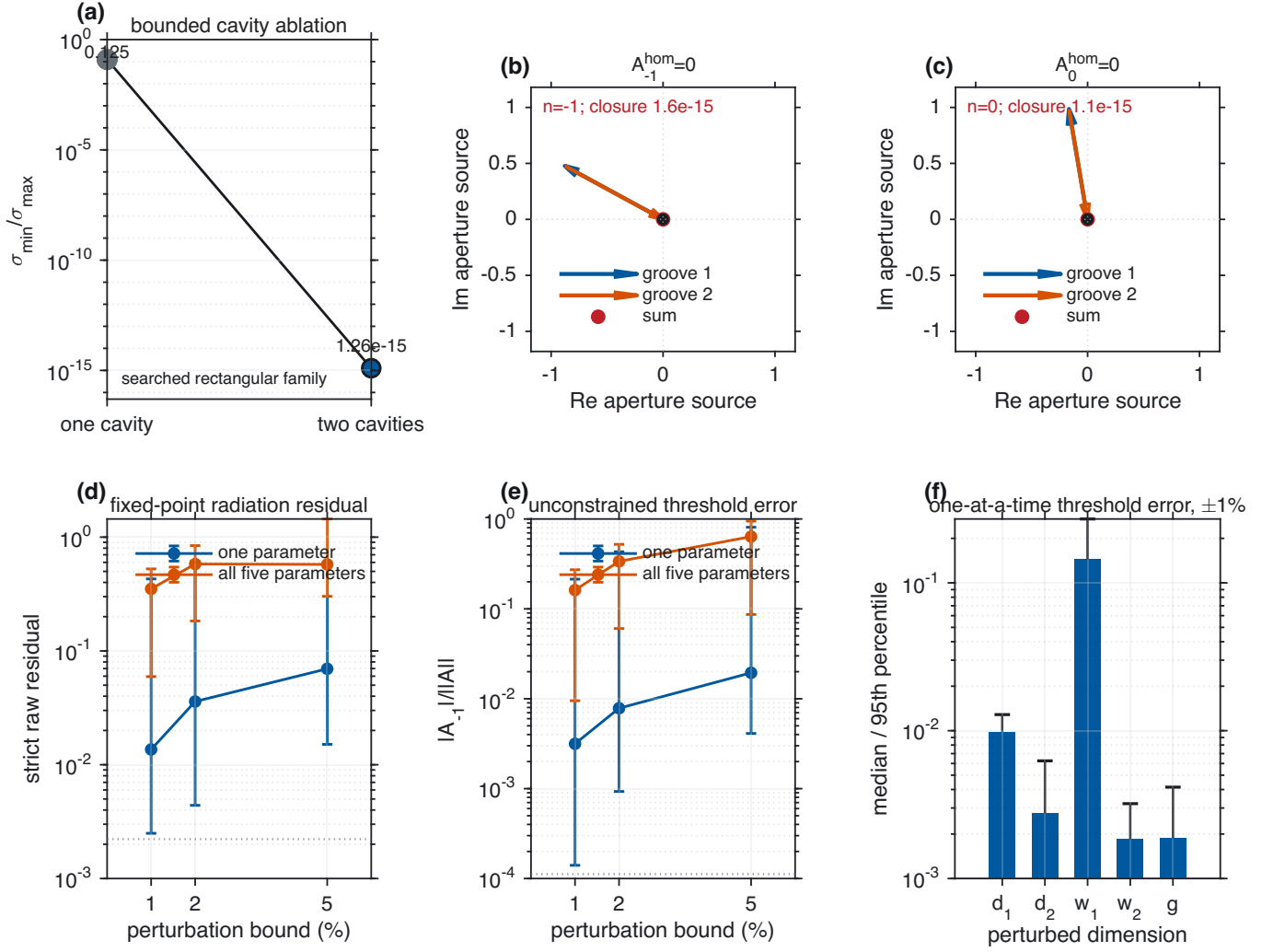


FIG. 4. Cavity degrees of freedom and fixed-point sensitivity. (a) Best single-large-cavity result in the searched rectangular family versus the two-groove root. (b,c) Separate aperture-source phasor closures for the grazing $n = -1$ and propagating $n = 0$ constraints. (d,e) Medians and 5–95% intervals of the strict raw residual and the unconstrained threshold-amplitude error for one-parameter and joint perturbations. The dotted line is the unperturbed (185, 23) numerical floor. (f) One-at-a-time $\pm 1\%$ sensitivity; bars are medians and whiskers are 95th percentiles.

We have thus identified a strict acoustic Rayleigh-threshold homogeneous null: the selected diffraction channel reaches grazing while its pressure amplitude and the remaining finite-flux amplitude vanish. The full operator, same-geometry pole, reoptimized root sequence, and exterior norm give complementary tests of this statement. The model assumes an infinite rigid surface, a lossless nondispersive fluid, and exact geometry. Finite-array leakage, viscothermal loss, elastic compliance, source aperture, and an independent full-wave calculation remain necessary before making an experimental or device-performance claim.

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